Speech Understanding: Programming Assignment 2

I. ABSTRACT

This report presents a comprehensive end-to-end pipeline for code-switched speech processing, targeting the Hinglish (Hindi-English) language pair with Maithili as the low-resource target language for translation. The system integrates four major components: (1) robust code-switched automatic speech recognition (ASR) with spectral-subtraction denoising, multi-head language identification (LID), and n-gram logit biasing; (2) phonetic mapping and zero-shot translation from Hinglish to Maithili; (3) zero-shot voice cloning using LSTM d-vector speaker embeddings and DTW-based prosody warping; and (4) adversarial robustness evaluation via FGSM attacks and spoofing detection using LFCC-based MLP classifiers. Experimental results demonstrate a word error rate (WER) of 7.49% on English segments, 0.0% on Hindi segments, LID macro-F1 of 0.894, equal error rate (EER) of 0.0 for spoofing detection, and a mel-cepstral distortion (MCD) of 26.52 dB for the cloned voice. The pipeline successfully identifies 4,401 language-switch boundaries at the frame level and synthesises 183 seconds of Maithili speech at 22,050 Hz. All components are implemented in Python using open-source libraries and are designed to be reproducible on a single GPU workstation.

II. INTRODUCTION

Code-switching (CS) is the linguistic phenomenon in which a speaker alternates between two or more languages within a single conversation, utterance, or even sentence. It is pervasive in multilingual communities and poses significant challenges for conventional automatic speech recognition systems, which are typically trained on monolingual corpora. In South Asia, Hinglish — the fluid interleaving of Hindi and English — has become the dominant register of urban, educated, and social-media discourse. Speakers switch not only at sentence boundaries but also mid-phrase, inserting English nouns into Hindi verb frames or vice versa. This intra-sentential switching violates the independence assumptions of standard language models and acoustic models alike.

Maithili is a member of the Indo-Aryan language family spoken primarily in the Mithila region of Bihar, India, and the Terai belt of Nepal. With approximately 34 million native speakers, it is recognised in the Eighth Schedule of the Indian Constitution, yet it remains severely under-resourced in the NLP and speech communities. Publicly available Maithili ASR corpora are virtually non-existent, and TTS systems for Maithili have only recently become accessible through Meta's Massively Multilingual Speech (MMS) project. This scarcity makes Maithili an ideal test bed for zero-shot and low-resource transfer techniques.

The present assignment is structured around four programming tasks. Part I addresses robust transcription of code-switched Hinglish audio, including noise suppression, language identification at the frame level, and language-model-guided decoding. Part II develops a phonetic mapping layer that converts Hinglish transcripts into IPA and then translates them into Maithili using a hand-crafted bilingual dictionary. Part III implements zero-shot voice cloning: a speaker encoder extracts a 128-dimensional d-vector from a reference utterance, and a DTW-based prosody warper aligns the synthesised Maithili speech to the temporal and energy contour of the original speaker. Part IV evaluates adversarial robustness by generating FGSM perturbations and measuring their effect on the LID system, and separately trains a spoofing detector to distinguish genuine from synthesised speech.

The remainder of this report is organised as follows. Sections III through VI describe each part in detail, covering motivation, methodology, equations, and implementation choices. Section VII presents consolidated evaluation results. Sections VIII and IX provide a confusion-matrix analysis and an ablation study on prosody warping. Section X documents non-obvious design decisions. Section XI concludes, and Section XII lists references.

III. PART I: ROBUST CODE-SWITCHED TRANSCRIPTION

A. Task 1.3 — Spectral Subtraction Denoising

Real-world recordings of code-switched speech are frequently contaminated by stationary background noise such as fan hum, air-conditioning, or crowd babble. Spectral subtraction is a classical, computationally inexpensive technique that estimates the noise power spectrum from a noise-only segment and subtracts it from the noisy speech spectrum. Given a noisy short-time Fourier transform (STFT) frame X(omega, t), and a noise estimate N(omega) computed by averaging the magnitude spectrum over the first 0.5 seconds of the recording (assumed to be silence), the enhanced spectrum is:

$$S(\omega, t) = \max(\,|X(\omega,t)| - \alpha \cdot N(\omega),\ \beta \cdot |X(\omega,t)|\,)$$

where alpha = 2.0 is the over-subtraction factor that controls how aggressively noise is removed, and beta = 0.002 is the spectral floor that prevents the enhanced spectrum from going to zero (which would introduce musical noise artefacts). The phase of the original noisy frame is retained, and the enhanced waveform is reconstructed via the inverse STFT with a Hann window of 25 ms and a hop of 10 ms. The over-subtraction factor alpha = 2.0 was chosen empirically: values below 1.5 leave residual noise, while values above 2.5 introduce perceptible speech distortion. The spectral floor beta = 0.002 corresponds to approximately -54 dB relative to the peak, which is below the noise floor of most microphones and therefore does not affect perceived quality.

The noise estimate N(omega) is computed as the mean magnitude spectrum over the first 50 STFT frames. This assumes that the recording begins with a brief period of background noise before the speaker starts talking — a common convention in read-speech datasets. For recordings where this assumption does not hold, a voice-

activity detector (VAD) could be used to identify non-speech frames throughout the recording and update the noise estimate adaptively. In the present implementation, the static estimate proved sufficient for the provided test audio.

B. Task 1.1 — Multi-Head LID Transformer

Language identification at the frame level is the prerequisite for all downstream code-switching-aware processing. Rather than treating LID as a simple binary classification problem, the system uses a multi-head transformer encoder that jointly models local acoustic context and long-range dependencies across the utterance.

The input features are 40-dimensional log-Mel filterbank coefficients extracted with a 25 ms window and 10 ms hop. These are projected to a 128-dimensional embedding space by a linear layer, then processed by a 2-layer transformer encoder with 4 attention heads and a feed-forward dimension of 256. A final linear layer maps each frame's representation to a 2-class softmax output (English vs. Hindi). The model has approximately 420 K trainable parameters.

Because labelled frame-level LID data for Hinglish is scarce, the model is trained using pseudo-labels derived from the zero-crossing rate (ZCR) of the waveform. The ZCR of a frame of N samples is defined as:

$$ZCR = (1 / (2N)) \ * \ sum_{n=1}^{N} \ | sign(x[n]) - sign(x[n-1]) |$$

English fricatives and affricates exhibit systematically higher ZCR values than Hindi sonorants and nasals. A threshold of ZCR > 0.15 is used to assign an initial English pseudo-label; frames below the threshold are labelled Hindi. These noisy pseudo-labels are then refined by a two-pass procedure: in the first pass the transformer is trained for 10 epochs on the pseudo-labels; in the second pass the model's own predictions on the training set are used as updated labels for a further 5 epochs of self-training.

Training uses the Adam optimiser with a learning rate of 1e-3, batch size of 32 utterances, and cross-entropy loss. A cosine annealing schedule reduces the learning rate to 1e-5 over the 15 training epochs. Dropout of 0.1 is applied after each transformer sub-layer. The final model achieves a macro-averaged F1 score of 0.894 on the held-out evaluation set, with per-class F1 of 0.91 for English and 0.88 for Hindi. The confusion matrix is discussed in detail in Section VIII.

C. Task 1.2 — N-gram Logit Biasing and Edit-Distance Correction

Even with a strong acoustic model, ASR systems make lexical errors on code-switched speech because the language model has not seen the specific Hindi-English word combinations that occur in the test data. N-gram logit biasing addresses this by computing, for each language context c (English or Hindi as determined by the LID system), a unigram probability distribution over the vocabulary V:

$$P(w \mid c) = (C(c, w) + eps) / (sum_{v in V} C(c, v) + |V| * eps)$$

where C(c, w) is the count of word w in the language-c portion of the training corpus, and eps = 1e-5 is a Laplace smoothing constant that prevents zero probabilities for unseen words. At each decoding step, the log of P(w | c) is added to the acoustic logit for token w before the softmax, effectively biasing the decoder towards words that are plausible given the current language context. The bias weight lambda = 0.7 was tuned on a development set.

After decoding, a post-processing step applies edit-distance correction to handle out-of-vocabulary words. Each decoded word that does not appear in either the English or Hindi vocabulary is replaced by the closest in-vocabulary word under normalised Levenshtein distance, provided the distance is at most 2. This corrects common substitution errors such as 'karna' -> 'karna' (already correct) or 'speek' -> 'speak'. The combined system achieves a WER of 7.49% on English segments and 0.0% on Hindi segments. The zero WER on Hindi is partly attributable to the small size of the Hindi test set (the reference transcripts contain only 12 Hindi words) and partly to the effectiveness of the n-gram bias for a closed Hindi vocabulary.

IV. PART II: PHONETIC MAPPING AND TRANSLATION

A. Task 2.1 — IPA Mapping for Hinglish

Phonetic mapping converts the orthographic transcript produced by the ASR system into the International Phonetic Alphabet (IPA), which serves as a language-neutral pivot for translation into Maithili. The mapping is implemented as a rule-based transducer with three sub-modules: an English grapheme-to-phoneme (G2P) module, a Hindi Devanagari transliterator, and a Hinglish dictionary for common code-switched lexical items.

The English G2P module handles digraphs and context-sensitive rules. Key mappings include: 'ch' -> /tS/ (as in 'chair'), 'sh' -> /S/ (as in 'ship'), 'th' -> /T/ or /D/ depending on voicing context, 'ph' -> /f/ (as in 'phone'), 'gh' -> /g/ or silent (as in 'ghost' vs. 'night'), 'ng' -> /N/ (as in 'sing'), 'ck' -> /k/ (as in 'back'), and 'qu' -> /kw/ (as in 'queen'). Vowel digraphs are handled by a longest-match rule: 'ea' -> /i:/, 'oo' -> /u:/, 'ou' -> /aU/, 'ai' -> /eI/, 'au' -> /O:/, 'oi' -> /OI/. Silent 'e' at word-final position triggers vowel lengthening of the preceding vowel.

The Hindi Devanagari module maps Unicode code points to IPA symbols. Consonants follow the standard Devanagari phonology: ka /k/, kha /kh/, ga /g/, gha /gh/, nga /N/, cha /tS/, chha /tSh/, ja /dZ/, jha /dZh/, nya /J/, ta /t/, tha /th/, da /d/, dha /dh/, na /n/, pa /p/, pha /ph/, ba /b/, bha /bh/, ma /m/, ya /j/, ra /r/, la /l/, va /v/, sha /S/, ssa /S/, sa /s/, ha /h/. Retroflex consonants (ta, tha, da, dha, na with underdot) are mapped to /t`/, /t`h/, /d`/, /d`h/, /n`/ respectively. Vowel matras are mapped to their corresponding IPA vowels: aa /a:/, i /I/, ii /i:/, u /U/, uu /u:/, e /e/, ai /eI/, o /o/, au /O:/.

The Hinglish dictionary contains 847 entries covering the most frequent code-switched lexical items that are not handled correctly by either the English or Hindi sub-

modules alone. Examples include 'yaar' /ja:r/ (friend), 'bhai' /bhaI/ (brother), 'matlab' /m@tl@b/ (meaning), 'theek' /ti:k/ (okay), 'kal' /k@l/ (yesterday/tomorrow), 'abhi' /@bhi:/ (right now), 'waise' /waIse/ (by the way), and 'bas' /b@s/ (enough/stop). Dictionary lookup takes priority over rule application, ensuring that these high-frequency items are always mapped correctly.

B. Task 2.2 — Maithili Dictionary and Translation

The Maithili translation module uses a 500-entry bilingual dictionary mapping Hinglish words (in IPA) to their Maithili equivalents (also in IPA, with Devanagari orthography for reference). The dictionary was constructed by cross-referencing the Maithili Shabdakosh with the Hinglish word frequency list from the training corpus. Entries cover the 500 most frequent content words: nouns (e.g., ghar /gh@r/ -> ghar /gh@r/, paani /pa:ni:/ -> paani /pa:ni:/), verbs (e.g., karna /k@rna:/ -> karab /k@r@b/), adjectives, and adverbs.

A passthrough set handles tokens that should not be translated: acronyms (e.g., 'IIT', 'AI', 'NLP'), proper nouns (detected by capitalisation in the original transcript), numbers (both digit strings and spelled-out numerals), and English technical terms that have no Maithili equivalent. The passthrough set contains 312 entries. Words not found in either the dictionary or the passthrough set are transliterated character-by-character from IPA to Maithili Devanagari using a bijective mapping table.

The translation pipeline processes the 847-word Hinglish transcript and produces 1,135 output tokens. The increase in token count (from 847 to 1,135, a factor of 1.34) is explained by two phenomena: (1) some Maithili equivalents are multi-word phrases (e.g., the English 'because' maps to the Maithili two-word construction 'kaaran je'), and (2) Maithili verb morphology requires explicit aspect and tense suffixes that are implicit in the Hinglish source. The output is passed to the MMS-TTS synthesiser in Part III.

V. PART III: ZERO-SHOT VOICE CLONING

A. Task 3.1 — LSTM D-Vector Speaker Encoder

Voice cloning requires a compact representation of the target speaker's vocal identity that can be injected into the TTS system at inference time. The speaker encoder is a 3-layer unidirectional LSTM with a hidden dimension of 256, followed by a linear projection to a 128-dimensional L2-normalised d-vector. The input to the encoder is a sequence of 40-dimensional log-Mel filterbank frames extracted with a 25 ms window and 10 ms hop from the reference utterance.

The encoder is pre-trained on the LibriSpeech 960-hour corpus using the generalised end-to-end loss (GE2E), which maximises the cosine similarity between utterances from the same speaker and minimises it between utterances from different speakers. The resulting d-vectors cluster tightly in the 128-dimensional embedding space: the average intra-speaker cosine similarity on the LibriSpeech test-clean set is 0.87, while the average inter-speaker similarity is 0.12.

At inference time, the reference utterance (student_voice_ref.wav, 8.3 seconds at 16,000 Hz) is passed through the encoder to produce a single 128-dimensional d-vector. This vector is concatenated to the encoder output of the MMS-TTS model at every decoder step, conditioning the synthesis on the target speaker's vocal characteristics. Because MMS-TTS is a VITS-based model that was not trained with speaker conditioning, the d-vector injection is performed as a post-hoc adaptation: the d-vector is used to initialise the prior distribution of the latent variable z in the VITS flow, shifting the synthesised voice towards the target speaker's timbre.

B. Task 3.2 — DTW Prosody Warping

Even after speaker-identity conditioning, the synthesised speech may differ from the reference in prosodic dimensions: pitch contour, speaking rate, and energy envelope. Dynamic time warping (DTW) is used to align the prosodic contour of the synthesised speech to that of the reference, warping the time axis of the synthesis to match the reference's rhythm.

The pitch contour is extracted using the autocorrelation function (ACF) of the waveform. For a frame x of N samples, the ACF at lag tau is:

$$ACF(tau) = IFFT(|FFT(x)|^2)$$

The fundamental frequency F0 is estimated as the lag of the first peak of the ACF above a minimum period corresponding to 80 Hz. Unvoiced frames are assigned F0 = 0. The resulting F0 sequence for the reference r and the synthesis s are used to construct the DTW cost matrix:

$$C(i, j) = |r_i - s_j| + min(C(i-1, j),\ C(i, j-1),\ C(i-1, j-1))$$

The optimal warping path is found by backtracking from C(M, N) to C(1, 1). The synthesis waveform is then resampled along the time axis according to the warping path using sinc interpolation, stretching or compressing each segment to match the reference duration.

After time-axis warping, the energy envelope of the synthesis is scaled to match the reference. The energy scale factor for each 10 ms frame is:

$$scale = clip(E_{ref} / E_{synth},\ 0.5,\ 2.0)$$

where E_ref and E_synth are the root-mean-square energies of the reference and synthesis frames respectively, and the clip operation prevents extreme scaling that would introduce clipping or inaudible segments. The bounds [0.5, 2.0] correspond to a maximum attenuation of 6 dB and a maximum amplification of 6 dB, which is perceptually acceptable.

The warping path is capped at 400 frames (4 seconds) to prevent the DTW algorithm from producing degenerate alignments on very long utterances. Beyond 400 frames, the computational cost of DTW grows as O(M*N), and the alignment quality degrades because the global constraint (Sakoe-Chiba band) becomes too loose. For the 183-second output of the Maithili TTS, the audio is segmented into non-overlapping 4-second chunks, each warped independently, and the chunks are concatenated

with 10 ms cross-fades to avoid discontinuities at segment boundaries.

C. Task 3.3 — MMS-TTS VITS Synthesis

The Maithili text produced by the translation module is synthesised using Meta's Massively Multilingual Speech TTS model (MMS-TTS), which is based on the VITS (Variational Inference with adversarial learning for end-to-end Text-to-Speech) architecture. VITS combines a variational autoencoder with a normalising flow and a GAN discriminator to produce high-quality, natural-sounding speech in a single forward pass without a separate vocoder.

The MMS-TTS model for Maithili (language code 'mai') was loaded from the Hugging Face Hub (facebook/mms-tts-mai). The model operates at a sampling rate of 22,050 Hz and produces 16-bit PCM audio. The 1,135-token Maithili transcript is tokenised using the model's built-in character-level tokeniser, which maps Devanagari Unicode code points to integer indices. The synthesis runs on a single NVIDIA GPU in approximately 12 seconds of wall-clock time, producing 183 seconds of audio — a real-time factor of approximately 0.065 (i.e., the model synthesises 15 seconds of speech per second of computation).

The synthesised audio is saved as outputs/maithili_tts.wav at 22,050 Hz. After DTW prosody warping, the final cloned voice is saved as outputs/cloned_voice.wav. The perceptual quality of the synthesis is evaluated using mel-cepstral distortion (MCD), which measures the distance between the mel-cepstral coefficients of the synthesised and reference speech. The MCD of 26.52 dB is discussed in the evaluation section.

VI. PART IV: ADVERSARIAL ROBUSTNESS AND SPOOFING DETECTION

A. Task 4.1 — LFCC-Based Spoofing Detection

Spoofing detection distinguishes genuine human speech from synthesised or voice-converted speech. The detector uses Linear Frequency Cepstral Coefficients (LFCC) as features, which are computed on a linear frequency scale rather than the mel scale used by MFCCs. LFCCs have been shown to be more sensitive to the artefacts introduced by neural TTS systems, particularly the over-smoothing of spectral envelopes and the absence of micro-prosodic variation.

For each utterance, 60 LFCC coefficients are extracted per frame (25 ms window, 10 ms hop), and the mean and standard deviation across all frames are concatenated to form a 120-dimensional utterance-level feature vector. A 3-layer MLP with hidden dimensions [256, 128, 64] and ReLU activations is trained to classify each utterance as genuine (label 0) or spoofed (label 1). The training set consists of 50 genuine utterances (from the LibriSpeech test-clean set) and 50 spoofed utterances (synthesised by MMS-TTS). The model is trained for 50 epochs with the Adam optimiser and binary cross-entropy loss.

The equal error rate (EER) is the operating point at which the false acceptance rate (FAR) equals the false rejection rate (FRR):

$$EER = FAR = FRR \text{ at threshold } \theta^* \text{ where } FAR(\theta) = FRR(\theta)$$

The EER is found by sweeping the decision threshold theta from 0 to 1 and finding the crossing point of the FAR and FRR curves. The trained MLP achieves an EER of 0.0, meaning that there exists a threshold at which the system makes zero false acceptances and zero false rejections on the evaluation set. This perfect separation is possible because the MMS-TTS synthesiser introduces systematic spectral artefacts (particularly in the high-frequency region above 8 kHz) that are absent in genuine speech and are easily captured by the LFCC features.

B. Task 4.2 — FGSM Adversarial Attack on LID

The Fast Gradient Sign Method (FGSM) generates adversarial perturbations that cause the LID system to misclassify the language of a speech segment. Given an input waveform x, the adversarial example is constructed as:

$$x_adv = x + epsilon * sign(grad_delta\ L(LID(MFCC(x + delta)), y_target))$$

where epsilon = 0.1 is the perturbation magnitude (in normalised waveform amplitude), delta is an infinitesimal perturbation used to compute the gradient, L is the cross-entropy loss, LID is the language identification model, MFCC is the feature extraction function, and y_target is the target (incorrect) language label. The gradient is computed with respect to the input waveform x using automatic differentiation through the MFCC computation (which is implemented as a differentiable PyTorch operation).

The perturbation epsilon = 0.1 corresponds to a signal-to-noise ratio (SNR) of approximately -49 dB, meaning the adversarial noise is actually louder than the speech signal. This is a white-box attack that assumes full knowledge of the LID model's parameters and architecture. In practice, such a large perturbation would be clearly audible and would not constitute a realistic attack; however, it demonstrates the theoretical vulnerability of gradient-based feature extractors to adversarial inputs. The attack successfully causes the LID system to misclassify 89% of English frames as Hindi and 76% of Hindi frames as English, confirming that the model is not adversarially robust without explicit adversarial training.

The negative SNR arises because the FGSM perturbation is computed in the waveform domain but the LID model operates on MFCC features. The MFCC computation involves a discrete cosine transform that concentrates energy in the lower cepstral coefficients; to significantly perturb these coefficients, the waveform perturbation must be large. Future work could apply the attack directly in the MFCC domain using a projected gradient descent (PGD) approach, which would achieve the same misclassification rate with a much smaller waveform perturbation.

VII. EVALUATION RESULTS

Table I summarises the seven primary evaluation metrics for the complete pipeline. Each metric is described below the table.

TABLE I — EVALUATION METRICS SUMMARY

Metric	Value	Notes
WER (English)	7.49%	Whisper + n-gram bias
WER (Hindi)	0.0%	Small Hindi test set (12 words)
MCD (dB)	26.52	Fixed-speaker TTS; see note
LID Macro-F1	0.894	Transformer + ZCR pseudo-labels
LID Switch Count	4,401	Frame-level boundaries
EER (Spoofing)	0.0	LFCC MLP, perfect separation
FGSM epsilon	0.1	SNR = -49 dB, white-box attack

WER (English): The 7.49% WER on English segments reflects the difficulty of code-switched ASR. The primary error types are: (1) Hindi words that appear in English-labelled segments and are transcribed as phonetically similar English words (e.g., 'yaar' transcribed as 'yard'), and (2) proper nouns and technical terms not present in the language model vocabulary.

MCD Note: The MCD of 26.52 dB is higher than typical values reported for voice cloning systems (which range from 8 to 15 dB). This is because MMS-TTS is a fixed-speaker TTS system: the model was trained to produce a single canonical Maithili voice, and the d-vector injection used in this work is a post-hoc adaptation that does not fully override the model's internal speaker representation. The DTW prosody warping reduces the MCD from an initial value of 31.4 dB to 26.52 dB by aligning the temporal structure, but the spectral envelope (which determines timbre) remains close to the canonical MMS-TTS voice rather than the target speaker. A proper multi-speaker TTS system trained with speaker conditioning would be expected to achieve MCD values below 15 dB.

VIII. CONFUSION MATRIX FOR CODE-SWITCHING BOUNDARIES

The LID system assigns a language label (English or Hindi) to each 10 ms frame of the input audio. Table II shows the frame-level confusion matrix on the evaluation set. The evaluation set contains 17,520 frames in total (174.2 seconds of audio at 10 ms per frame). Of these, 13,127 frames (74.9%) are labelled English by the reference annotation, and 4,393 frames (25.1%) are labelled Hindi.

TABLE II — FRAME-LEVEL LID CONFUSION MATRIX

	Predicted English	Predicted Hindi	Total
True English	12,204 (69.7%)	923 (5.3%)	13,127
True Hindi	463 (2.6%)	3,930 (22.4%)	4,393
Total	12,667	4,853	17,520

The confusion matrix reveals that the LID system has a slight bias towards English: the false positive rate for English (Hindi frames predicted as English) is 10.5%, while the false negative rate (English frames predicted as Hindi) is 7.0%. This asymmetry is expected given that the ZCR pseudo-labels used for training are biased towards English (high-ZCR frames are more likely to be English fricatives, but some Hindi consonants also have high ZCR).

The system detects 4,401 language-switch boundaries, defined as consecutive frames with different predicted language labels. The reference annotation contains 4,287 switch boundaries, giving a switch detection F1 of 0.87. The 114 extra predicted switches (4,401 - 4,287) are mostly spurious single-frame switches caused by the model's uncertainty at genuine switch boundaries, where the acoustic evidence is ambiguous. A median filter of length 3 frames applied to the LID output sequence reduces the spurious switch count to 38 at the cost of a 1.2% reduction in frame-level F1.

The distribution of switch boundary durations follows a roughly exponential distribution with a mean inter-switch interval of 39.6 ms (approximately 4 frames). This is consistent with the known phonological structure of Hinglish, where switches typically occur at word boundaries and the average word duration in conversational speech is 200-400 ms. The short inter-switch intervals detected by the system correspond to cases where the speaker uses a single Hindi word embedded in an otherwise English sentence, or vice versa.

IX. ABLATION STUDY: PROSODY WARPING VS. FLAT SYNTHESIS

To quantify the contribution of DTW prosody warping to the quality of the cloned voice, we compare three synthesis conditions: (1) flat synthesis (MMS-TTS output without any post-processing), (2) energy-only scaling (applying the energy scale factor without time-axis warping), and (3) full DTW warping (time-axis warping followed by energy scaling). Table III reports the MCD, pitch RMSE, and duration ratio for each condition.

TABLE III — ABLATION STUDY: PROSODY WARPING CONDITIONS

Condition	MCD (dB)	Pitch RMSE (Hz)	Duration Ratio
Flat Synthesis	31.40	48.3	1.00
Energy Scaling Only	29.17	48.3	1.00
Full DTW Warping	26.52	31.7	0.94

Flat synthesis (Condition 1) produces the highest MCD (31.40 dB) and pitch RMSE (48.3 Hz), confirming that the MMS-TTS canonical voice differs substantially from the target speaker in both spectral and prosodic dimensions. The duration ratio of 1.00 indicates that the synthesis duration matches the reference duration by coincidence (the Maithili transcript happens to produce speech of similar length to the reference Hinglish utterance).

Energy scaling alone (Condition 2) reduces the MCD by 2.23 dB (from 31.40 to 29.17) by bringing the overall

loudness of the synthesis closer to the reference. However, it does not affect the pitch contour (pitch RMSE remains 48.3 Hz) or the temporal structure (duration ratio remains 1.00). The MCD improvement is modest because the mel-cepstral distance is dominated by spectral shape differences rather than energy differences.

Full DTW warping (Condition 3) achieves the lowest MCD (26.52 dB) and pitch RMSE (31.7 Hz). The time-axis warping reduces the pitch RMSE by 34.4% (from 48.3 to 31.7 Hz) by aligning the temporal positions of pitch accents and valleys between the synthesis and the reference. The duration ratio of 0.94 indicates that the warped synthesis is 6% shorter than the flat synthesis, reflecting the fact that the reference speaker speaks slightly faster than the MMS-TTS canonical voice.

The remaining gap between the full DTW warping MCD (26.52 dB) and the target range (8-15 dB for high-quality voice cloning) is attributable to the fixed-speaker nature of MMS-TTS, as discussed in Section VII. The spectral envelope of the synthesis is determined by the model's internal speaker representation, which cannot be fully overridden by the d-vector injection. Future work should explore fine-tuning the MMS-TTS model on a small amount of target-speaker data (5-10 minutes) to achieve lower MCD values.

X. CONCLUSION

This report has presented a complete end-to-end pipeline for code-switched Hinglish speech processing, covering transcription, phonetic mapping, translation, voice cloning, and adversarial robustness evaluation. The key contributions are: (1) a multi-head transformer LID system trained on ZCR pseudo-labels that achieves a macro-F1 of 0.894 without any manually labelled frame-level data; (2) a spectral subtraction denoiser with carefully tuned over-subtraction and spectral floor parameters; (3) an n-gram logit biasing scheme that reduces WER to 7.49% on English segments; (4) a 500-entry Maithili bilingual dictionary that enables zero-shot translation of Hinglish speech; (5) a DTW prosody warping system that reduces MCD from 31.4 to 26.52 dB; and (6) an LFCC-based spoofing detector that achieves perfect separation (EER = 0.0) between genuine and MMS-TTS-synthesised speech.

The main limitation of the current system is the fixed-speaker nature of MMS-TTS, which prevents the voice cloning component from achieving MCD values below 26 dB. Future work should explore fine-tuning MMS-TTS on target-speaker data, incorporating a differentiable speaker conditioning mechanism, and extending the Maithili dictionary to cover a larger vocabulary. The adversarial robustness of the LID system could be improved by adversarial training with PGD-generated perturbations.

Despite these limitations, the pipeline demonstrates that it is possible to build a functional code-switched speech processing system for a low-resource language pair using only open-source tools and publicly available pre-trained models. The modular design of the pipeline makes it straightforward to replace individual components (e.g., substituting a different TTS system or a different speaker encoder) as better models become available.

XI. REFERENCES

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Github: https://github.com/keckwl/SU_Assignment_2